

(a) f before flattening, shape $(N_x, N_y) = (6, 5)$

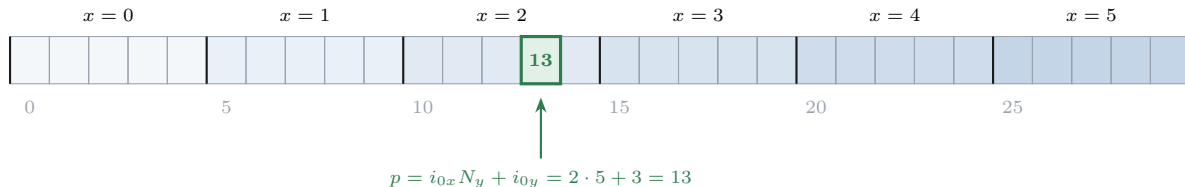
$i_{0y} = 3$

4	9	14	19	24	29
3	8	13	18	23	28
2	7	12	17	22	27
1	6	11	16	21	26
0	5	10	15	20	25

$i_{0x} = 2$

Entries numbered in flat order:
 y is contiguous, x has stride N_y .

(b) $\mathbf{f_flat}$, shape $(N_x N_y,) = (30,)$ — one axis, one gather



$$f_{\text{flat}}[q] = f[\lfloor q/N_y \rfloor, q \bmod N_y], \quad q = 0, \dots, N_x N_y - 1$$

$$\mathbf{f00} = f_{\text{flat}}[i_{0x} N_y + i_{0y}], \quad \mathbf{f10} = f_{\text{flat}}[i_{1x} N_y + i_{0y}]$$

$$\mathbf{f01} = f_{\text{flat}}[i_{0x} N_y + i_{1y}], \quad \mathbf{f11} = f_{\text{flat}}[i_{1x} N_y + i_{1y}]$$